

The Conditional-to-Interrogative Cycle: Empirical Evidence from German Varieties¹

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1 INTRODUCTION

It is a well-known fact that in many languages the complementizers in conditional and polar interrogative subordinate clauses are homonymous. The examples (1) to (3) illustrate this fact for Indo-European languages, but it is also found in languages from other language families like Gorwaa (Cushitic) (4), Maltese (Semitic) (5), or Malagasy (Austronesian) (6).

- (1) a. You can go **if** you want. (English)
b. I was wondering **if** you want to go.
- (2) Ik gao even vraogen **as** het heur wel past **as** we kommen
I go briefly ask as it her well fits as we come
'I'm just going to ask her, whether it would suit her if we come.'
(Stellingwerf dialect (Friesland), Bloemhoff 2004: 218)
- (3) a. **Se** vuoi, puoi andare. (Italian)
'If you want, you can go'
b. Mi stavo chiedendo **se** vuoi andare ancora.

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‘I was wondering if you want to go again.’

- (4) a. Tlakweema **bir** tleehhít tlakweemaw-ók inki/iyá’
 evil COND:O.N make:2SG evil-2SG.POSS do.again:3N
 ‘If you do evil, your evil will return’ (Gorwaa, Edward & Harvey 2013)
- b. A xuuakáng gidaábá daaqaay **bir** ni xáy
 AUX know:1SG:NEG COMP boys COND VEN come:3M
 ‘I don’t know if the boys will come’ (Gorwaa, Melle Groen, p.c.)
- (5) a. Is-surmast saqsieni (**jek**k) il-gimgha d-diehla hux se tkun vaganza
 the-headmaster asked:3M.SG:PRN.1SG. (if) the-week the-entering:F.SG
 NEG:FUT:be:3F.SG holiday
 ‘The headmaster asked me whether next week was a holiday’
 (Maltese, Azzopardi-Alexander & Borg 1997: 23)
- b. **Jekk** jitla’ s-shab, taghmel ix-xita
 if goes.up:3M.SG the-clouds does:3F.SG the-rain
 ‘If clouds form, it will rain.’
 (Maltese, Azzopardi-Alexander & Borg 1997: 38)
- (6) a. Zahao **raha** marina izany
 Look:MP if true DEM:DIST
 ‘Check whether this is true.’
- b. **raha** te-ho sambatra hianao, tiavo ny hatsaram-panahy
 if want-IRR happy 2.SG.TOP, love.IMP ART good-heartedness
 ‘If you want to be happy, love kindness.’
 (Malagasy, Malzac 1926: 761)

Even though this ‘conditional-interrogative link’ (Starr 2014) is so pervasive, it has not been widely discussed from a diachronic viewpoint so far. In the present paper

we will present data from historical and dialectal German that shows that the conditional function is older than the interrogative one. At first sight, German does not seem to be a paradigmatic test bed, as the modern Standard variety shows two different complementizers: *ob*, the cognate of English *if*, in interrogatives and *wenn* in conditionals:

- (7) a. Du kannst gehen, **wenn** du willst.
 ‘You can go if you want.’
 b. Ich frage mich, **ob** du gehen willst.
 ‘I am wondering if you want to go.’

Conditional *ob* only occurs in very special clause types, for example in alternative irrelevance conditionals (d’Avis 2004):

- (8) **Ob** es regnet oder schneit: Wir gehen wandern!
 ‘Whether it is raining or snowing: We are going hiking!’

Wenn, by contrast, cannot be used as an interrogative complementizer in Standard German. This use, however, is attested in Saurian, a German dialect of northern Italy (*ben* = *wenn*; cf. section 3 for details):

- (9) Ar otmi gevörschet **ben** der meiget helfnme de makina za richtan!
 He has-me asked when he may help-me the car to fix
 ‘He asked me if he could help me fix the car’

In the following we are going to discuss the uses of these and further complementizers in Old High German (section 2) and Saurian (section 3). Based on empirical evidence from these varieties, we will go into the diachronic pathways through which conditional complementizers take over interrogative functions in a cyclical manner (section 4). Section 5 discusses whether the reverse path (from

interrogative to conditional complementizer) is also attested. Section 6 concludes the paper.

2 CONDITIONAL AND INTERROGATIVE COMPLEMENTIZERS IN THE HISTORY OF GERMAN

In Old High German (OHG), the oldest attested stage of the German language, the most frequent conditional complementizer is *ibu/ubi/oba*, cf. the examples in (10).² Cognate or closely corresponding forms are attested in many Germanic varieties (e.g. Old Frisian *jef, jof, ef, of*; Old Dutch *of*; Old Saxon *ef, af, of*; Old Icelandic *if, ef*; Old Danish *æf, of*).

- (10) a. enti **ibu** daz in gro pa fallit inrestitagû Inu nimit iz dêr [...]

‘And if it (=the sheep) falls into a pit on the sabbath day, will he not lay

hold on it?’ (DDD-AD-Monsee_1.2 > MF_1_M.IV)
- b. **oba** thin ouga uuirdit luttar, thanne ist al thin lihhamo liohter

‘If your eye is clear, then your whole body is lighted’

(DDD-AD-Tatian_1.2 > T_Tat36)

OHG, in contrast to New High German (NHG), showed the ‘conditional-interrogative’ link. As the examples in (11) illustrate, indirect interrogatives were also introduced by *ibu/oba* and variants of these:

- (11) a. fona himile simblum sihit ubar parn manno, daz sehe,

from heaven always sees onto children men’s that see.SBJV.3SG

ibu ist farstantanti edo suahhanti cotan

if is understanding or seeking God.ACC

² In the following, unless otherwise indicated, the OHG examples were taken from the ReA corpus and searched for with ANNIS4.

‘God looks down from heaven on the children of man to see if there are any who understand or seek after God.’

(DDD-AD-Benediktiner_Regel_1.2 > B_7)

- b. nu chius dir fasto ze mir , **u**pe ih mih chere after dir

‘Now check me carefully to see if my path leads to you’

(DDD-AD-Kleinere_Althochdeutsche_Denkmäler_1.2 Ps138_Psalm138)

- c. Strîr uuîrdet . **ú**be éin mán urî sî

‘There will be a dispute whether a person is free.’

(DDD-AD-Z-Notker_Kleinere-Syllogismus_1.2 > N_Syl_1)

May etymological information help us to decide which function was primary, the conditional or the interrogative one? Unfortunately, the etymological origin of OHG *ibu/oba* is quite obscure. It has been hypothesized to go back to Germ. **ja-ba*, a reconstructed noun with the meaning ‘doubt’ (Klein 1971). And indeed we find nouns such as OHG *iba* ‘condition, stipulation, doubt’; Old Icelandic *if* ‘doubt, hesitation’; Swedish *jä* ‘exception, challenge’, but it seems equally plausible that these have developed secondarily from the conjunction the other way round.³ What is also puzzling are the varying forms (cf. the list in Lühr 2007: 39: *ibu, ibi, ipu, ubi, uba, ub(e), upi, upe, uoba, oba, ob(e), obo, aba, abo, avo*), and here in particular the alternation between *i* and *u/(o)* in the first syllable. Lühr (1976, 2007) dismisses previous explanations of the *u*-anlaut as change from *e* to *o* under weak stress (‘Schwachton’) or transposition of vowels because these types of changes are not attested in other cases in OHG. In her proposal the *u*- and *i*-forms have two different sources. She proposes that the *u*-anlaut in the *u*-forms is originally an interrogative particle (cf. Gothic *-u*, which was used in disjunctive and in indirect interrogatives). The Proto-Germanic form was probably provided with a *b*-particle in the second syllable (= **u-bē* from Pre-Proto-Germ. **u-b^heh*, cf. Lühr 1976: 82), built on forms such as **ia-bāi*, which was continued as *jabai* ‘if’ in Gothic (cf.

³ A reviewer points to parallels to this in present-day languages, e.g. English *ifs and buts*, much discussed in the literature on degrammaticalization (e.g. Norde 2001: 235).

below). The *i*-forms, by contrast, Lühr (2007) argues to originate in the negative form of the complementizer. She reconstructs a compound for with the negation particle *ni* for Proto Germanic, namely **ni-bō* ‘if not’, of which, in turn, positive counterparts were built through back-formation already in Proto-Germanic. Continuations of this are Goth. *niba* and *iba* and OHG *nibu* and *ibu*. Due to the drop of the initial *n-*, the originally negative connector acquired a non-negative meaning ‘if’. As mentioned, Lühr hypothesizes that there was a form **ja-bāi* in Proto-Germanic, where the first syllable probably goes back to a pronominal with a conditional meaning originally, cf. e.g. the Old Indic conjunction *yád*. The *b*-part was originally a strengthening particle meaning something like ‘really, indeed’.⁴ The Goths, however, must have reinterpreted the semantic composition of *jabai* and relocated the conditional interpretation to *-bai*, thereby treating *ja-* probably as the affirmative particle *ja* ‘yes, truly’. Evidence that it is *-bai* that is contributing the conditional meaning in Gothic comes from examples where it is used without *ja* as an enclitic particle:

- (12) [...]saei galaubeiþ du mis,
 whoever believes to me
 þauh ga-**ba**-daupniþ, libai
 though ga.PREF-BA.PART-believe.3SG.PRES.IND live.3SG.PRES.OPT
 ‘he that believes in me, though he may die, he shall live’
 (J 11, 25; from Lühr 2007, ex. 9; English glosses and translation added)

Thus, according to Lühr, there were originally two words/morphemes for conditional and interrogative meanings in Germanic: one with an *i*-anlaut (recruited

⁴ The *b*-part is attested with different vowels in OHG. While *-a* and *-o* can be weak-toned variants and the final *-e*, (e.g. in Notker’s work), already indicates the development into a schwa sound, the final *-u*, *-a* and *-i* are definitely old (Lühr 2007). Lühr hypothesizes them to be reflexes of case forms of a Proto-Indo-European pronominal stem **b^he/b^ho-* that has only been retained in adverbs and particles. The somewhat problematic *-bi* in OHG *ubi* could accordingly originate in a Pre-Proto-Germanic locative **b^heī* or *b^hi* or, alternatively, in a pre-OHG enclitic particle (**u-b-ī*, shortened to **u-b-i*) as witnessed in the Gothic enclitic relative particle *-ei* (Lühr 2007).

from the negation **ni*) and one with *u-*. If Lühr’s (2007) theory of the two different origins is correct, there would have been a development from a conditional into an interrogative complementizer in Germanic in the back-formed *i*-forms. Lühr (2007) points to semantic similarities that might have driven this change (cf. section 4 for further discussion). With the *u*-form, the situation is less clear: Lühr (2007) does not explicitly discuss the relative chronology of the two processes, namely the transference of the conditional meaning onto the *b*-part and the procliticization of the interrogative *u*-part in the continuations of **īa-bāī*, but mentions that there may have been a reciprocal interference between conditional and interrogative connectors. In her earlier work (Lühr 1976), she explicitly speculates that in a Pre-Proto-Germanic **u-b^heh* compound form, the *b*-particle was still a strengthener reinforcing the interrogative particle in the anlaut.

If we focus only on the evidence in OHG again and look at the two uses from a quantitative perspective, we notice the conditional use of *ibu/oba* outnumbers the interrogative one by far. The following table gives the figures for the OHG Tatian translation, for Otfrid’s Gospel book, for the ‘Minor OHG monuments’ and for Notker Labeo’s texts. The data were obtained through a study in ReA: We searched the relevant texts with the queries “lemma=/ibu/ & translation=/ob/ & #1=_#2”, “lemma=/ibu/ & translation=/wenn, falls/ & #1=_#2”, “lemma=/ibu/ & translation=/wenn/ & #1=_#2”.

<i>ibu/oba</i> etc.	# interrogative	# conditional
Tatian	9	167
Otfrid	12	84
Minor OHG documents	5	46
Notker’s works	13	165

Fig. 1: Numbers for interrogative and conditional *ibu/oba* in various OHG texts

The quantitative predominance of conditional *ibu/oba* over the interrogative one probably derives from the fact that there are more conditional relations to be expressed than indirect questions in these OHG sources. It does not provide us with any evidence for the relative chronology of the two complementizers. Both also only had minor competitors. In indirect interrogatives, OHG additionally witnesses the subordinator (*h*)*wedar* besides *ibu/oba*, a cognate of modern English *whether*, but this use is very infrequent, cf. (13a) for an early example from the Isidor translation and (13b) for a later attestation from Notker.⁵

- (13) a. Suohhemes auur uuir nu ziidh dhera christes chiburdi,
 huuedhar ir iu quhami odho uuir noh sculim sin quhemandes biidan
 ‘Let us now seek the time of Christ’s birth, whether he already came or we
 should still be waiting for his coming’
 (DDD-AD-Isidor_1.2 > I_DeFide_5)
- b. Uuér máhtî ian démo strîte chéden. **Uuéder** iz réht. álde únreht uuâre?
 ‘Who can say of the quarrel whether it is right or wrong’
 (DDD-AD-Notker Boethius-De_Consolatione_philosophiae_1.2)

The interrogative subordinator *wedar* evolved from the interrogative pronoun/determiner *wedar*, which had the meaning ‘which of the two’. In Old English we find more attestations of *hwæper* (also as a disjunctive question particle in independent interrogatives) and the development of the pronoun with the ‘which of the two’ reading into the subordinator with the disjunctive question reading can thus be sketched more straightforwardly (Walkden 2014, Eckardt & Walkden 2022, Axel-Tober & Beck 2024). Since the disjunctive-question use is not attested in

⁵ An anonymous reviewer asks if *wedar* is attested with the same types of matrix predicates as *ibu/oba*. Since there are only a couple of attestations, we cannot really come to definite conclusions. Besides *suohhen* and *quedan* as in (13) above, we find *wedar* after *firstān* (Tatian) and (*ne*) *wizzan*, in *zwīfale sīn* (Notker) or in a subject clause after the phrase *an iro iuditio stāt* (*uueder*) ... ‘in their judgement it is stated (whether)’ (Notker). – Example (13a) retains the flavor of ‘which of two’, though with explicit clausal disjuncts.

Gothic, it was probably not available in Proto-Germanic (Walkden 2014). In German, there are some sporadic attestations of it in MHG texts, but it later died out in this use. Neither in English nor in German nor in any other Germanic language did *whether*-forms develop a conditional interpretation.⁶

As to conditional competitors, *sō(sō)* is also attested as a conditional subordinator in OHG, often in a correlative structure with homonymous *so* functioning as a resumptive adverb in the consequent clause (Schmid 2023, § 209, Axel-Tober 2023):

- (14) **so** iz regenot, so nazscent te boumma
 SO (=if) it rains SO (=then) wet the trees
 ‘if it rains, the trees get wet.’
 (OHG proverbs; from cf. Schmid 2023, § 209)

Behaghel (1928: 350) also mentions that the OHG precursor of relative *wo*, OHG *so (h)wār(a) (so)* ‘wherever’, is attested in examples where it comes close to a conditional-like interpretation (cf. also section 4.2).

The precursor of our modern German standard conditional complementizer *wenn*, *sō (h)wanne (sō)*, primarily has a temporal interpretation ‘as soon as, as often as, whenever’ in OHG. Even in MHG, conditional uses of *swenne* (with umlaut) are not attested robustly and unambiguously, even though the meaning touches on the conditional multiple times (BMZ, lemma = *swenne*); cf. (15).

- (15) sît gewis, **swenn** ir uns komet, ir werdet hôh enpfangen
 be sure, whenever you to.us come, you become highly received
 ‘Be sure, whenever/when/if you come to us, you will get a high-ranking reception.’
 (Walth. 48,19., cited from BMZ; glosses and translation added)

⁶ *Whether* occurs in so-called alternative unconditionals (*Whether Elif comes to the party or not, it will be fun*) (Rawlins 2002). Cf. also ex. (8) above for a German alternative irrelevance conditional with *ob*.

It was not until the 15th century that *wenn* increasingly displaced the older *ob*-complementizer in the various types of conditional clauses. In this transitional period *ob* and *wenn* still competed with *so* and *wo* (Behaghel 1928: 348).

So, *wo* and *wenn* all also developed disjunctive readings in indirect interrogatives. However, these developments did not take place in Standard German, but only in certain dialects. In the following we will discuss data from Saurian to show the ‘conditional-interrogative’ link with two of the three complementizers (namely *wo* and *wenn*). In these cases, it is transparent that the conditional uses precede the interrogative one.

3 FROM CONDITIONAL TO INTERROGATIVE COMPLEMENTIZERS IN SAURIAN

3.1 SAURIS

Sauris (Zahre) is a South-Bavarian linguistic island in the Carnian Alps of Friuli, northeast Italy (Denison 1997: 68). It has been populated by people from East Tyrol and/or Carinthia since the late 13th century, a process which came from the “vicinity of Upper and Lower Tilliach, Kartitsch, and Maria Luggau” (Denison 1997: 69). Today, Sauris consists of the three main villages of Sauris di Sotto (Untertzahre/Dörf), Sauris di Sopra (Oberzahre/Plozn), Lateis (Latais) and the two settlements of La Maina (Ame Lataise) and Velt. Sauris is one of several German linguistic islands in Northern Italy (cf. fig. 2):



Fig. 2: Linguistic islands in Northern Italy (source: www.isolelinguistiche.it)

The inhabitants of Zahre were trilingual until the middle of the 20th century (cf. Denison, et al. 1968, 1994): they spoke Zahrish/Saurian within the family and in the village, Friulian with the inhabitants of the neighboring villages and the lowlands, as well as Italian (as a written language and in official dealings). After decades of decline in use until the 1980s, the proportion of Saurian speakers today is around 70% (Cattarin 1999/2000, quoted from Protto 2006); younger age groups increasingly have only passive knowledge, although this has recently been supported by school and church. Today, Zahre has 390 inhabitants (cf. <https://www.sauris.org/de/>).

The Saurian data presented in the following was collected in November 2017 in interviews with seven informants of various ages from the three main villages, using standardized questionnaires for syntactic surveys. The interviews were conducted by Ermenegildo Bidese (Trento), Alessandra Tomaselli (Verona) and Helmut Weiß (Frankfurt am Main) and the informants were given Italian sentences to translate into Saurian. AUG, ARM, ART, ERM, FER, GRA, and LAR are acronyms of our informants whom we sincerely thank for their invaluable help.

3.2 SAURIAN COMPLEMENTIZERS

Saurian (or Sauris German) is a South Bavarian variety which has its origin in Tyrolean, but also shows influences of the Carinthian. It is closely related to the language islands of Sappada (Bladen/Plodn) and Timau (Tischelwang/Tischlbong) (cf. Fig. 2), with which contacts have always existed. The Bavarian language island dialects in northern Italy have developed largely without contact to Standard German. They show special features at all linguistic levels, some of which were also caused by the influence of the surrounding Italian varieties, most others, however, are language-internal developments (cf. Bidese 2023 on Cimbrian). A phonological peculiarity of Saurian is that the fricative /v/ (<w>) has developed into the plosive /b/, i.e., *Winter* ‘winter’ is /binter/ and *Wasser* ‘water’ is /bosser/ (Schneider 1999: 10). This development also occurred in other Bavarian linguistic islands in northern Italy (for example, in Mòcheno, cf. Rowley 2003) and it is also occasionally found in Inner German dialects, but there it is limited to interrogative pronouns (Schirmunski 1962: 366).

Multiple complementizers are a syntactic peculiarity. Since Rizzi (1997), it has been common practice in syntax to split the C-domain into several separate functional projections. The so-called split-CP (cf. e.g., Radford 2004) includes projections for more traditional phi features such as ForceP (sentence mood) and FinP (finiteness features), but also information-structurally relevant projections for topic and focus; in Rizzi (2001) an extension by an interrogative phrase was proposed, cf. (16):

- (16) [ForceP [IntP [TopP [FocP [FinP ...

The split-CP hypothesis is assumed above all for Romance languages, for German (varieties), however, there is hardly any direct empirical evidence that the C domain is split into multiple projections. The restriction to one constituent in the prefield that applies to strict V2 languages even seems to speak against a split CP in

German.⁷ Saurian is now a German variety that provides clear evidence for the existence of a differentiated left periphery, because it uses multiple complementizers very systematically (cf. Weiß 2023). On the one hand, this can be observed with adverbial complementizers, which can be combined with *(a)s* ‘that’ (but do not have to be).⁸ For example, the temporal complementizer *be(i)n(e)* ‘when’ meaning ‘as’ (17) or ‘when’ (18) occurs with and without *(a)s* ‘that’:

- (17) A3. Quando il cacciatore si è mosso, la volpe è scappata

LAR: **Ben** ’s der jeger otsi geriert, der vuks ist intrunen

ARM: **Ben** der khjokhmon otsi geriert, der vuks ist intrunen

GRA: **Ben** ’s der khjokhmon otsi geriert, der vuks ist intrunen

As that the hunter has-himself moved, the fox is escaped

‘When the hunter moved, the fox escaped’

- (18) C4. Non incontro mai nessuno quando vado a correre (nel bosco)

LAR: I tue nie khans pageign **ben** i gei za lafan tschmörganz

FER: I tue nie khans pageign **beine as** i gea za lafan im bolde

ARM: I tue nie khans pageign im bolde **ben** i gea za lafan

ART: I pageigne nie khans **ben** i gea in bolt

ERM: I tue nie khans pageign **ben** i gea im bolde

I do never no-one meet when I go to run in.the forest

‘I never meet anyone when I go running in the forest’

Multiple complementizers are clear evidence for the existence of several functional projections in the left periphery. The most obvious structural analysis is that the

⁷ But see Volodina & Weiß (2010), Weiß (2011), Catasso (2015), or Speyer & Weiß (2018) for split-CP analyses of German.

⁸ The lexeme *ass* that occurs in Saurian instead of *dass* ‘that’ (as well as in some other German dialects) is traditionally traced back to *dass* (formed by the loss of the initial ‘d’). Axel-Tober & Brandner (acc.) offer an alternative explanation, according to which Germanic AS-type declarative complementizers go back to *als* ‘as’.

individual complementizers occupy different head positions, e.g. *be(i)n(e)* ‘when’ Force° and *(a)s* ‘that’ Fin°, cf. (19):

- (19) [ForceP *be(i)n(e)*] [IntP [TopP [FocP [FinP *(a)s* ...

In Saurian, conditional sentences are introduced with *be(i)n(e)*, but apparently without an additional *(a)s* ‘that’. In our data there are two tasks with conditional clauses. In both the additional second complementizer is missing in the translations, cf. (20) and (21):⁹

- (20) A8. Se farai un buon lavoro, ti aiuteremo volentieri
 LAR: **Ben** de barst ana scheana orbat mochn, tueberder gearn helfn
 FER: **Ben** de barst tuen ana gueta orbat, barberder gearn helfn
 ARM: **Ben** de tuest mochn ana gueta orbat, der barnder helfn gearn
 ART: **Ben** de tuest ana scheana orbat, tueberder gearn helfn
 GRA: **Ben** de mochest ana scheana orbat, tueberder gearn helfn
 If you will/do make a beautiful work, will/do-we-you help gladly
 ‘If you do a good job, we will be happy to help you’
- (21) B13 Se piove (neve) tutta la giornata, rimaniamo a casa
 LAR: **Ben** ’s regnt in gonze tokh, plaibeber dehame
 AUG: **Ben** ’s in gonze tokh regntis / tuets regn, plaibeber dehame
 ART: **Ben** ’s haite regnt in gonze tokh, plaibeber dehame
 ERM: **Ben** ’s tuet regn in gonze tokh, der plaibn dehame
 GRA: **Ben** ’s schnaibet in gonze tokh, plaibeber dehame
 If it rains the whole day, stay-we at.home
 ‘If it rains/snows all day, we stay at home’

⁹ Since we did not inquire whether an additional *(a)s* is possible or not, we can only observe that it does not appear in our data. Note that the ‘s’ occurring in (21) is not the complementizer, but the expletive subject of the weather verb *regnen* ‘rain’.

The complementizer *be(i)n(e)* ‘when’ thus introduces temporal and conditional clauses in Saurian (as in Standard German).

3.3 FROM CONDITIONAL TO INTERROGATIVE COMPLEMENTIZER

In Saurian, the complementizer *be(i)n(e)* has yet another function: indirect polar interrogatives are not introduced with *ob* ‘whether’ (as in Standard German and in most dialects), but with *be(i)n(e)*, whereby *(a)s* is also possible as second complementizer. In this interrogative function, some speakers also use *bo* (*'s*) lit. ‘where (that)’ (see below). In German dialects, the only other interrogative complementizer is *was*, lit. ‘what’, namely in Low German and Bavarian (cf. Weiß 2013: 778), cf. (22a, b):

- (22) a Ik weet nich, **wat** de Bodder al smolten is
 I know not, what the butter all melted is
 ‘I don't know if the butter has melted completely.’
 b. wer woäß, **wos** si d Dora vo mia no oamal fohrn loußt
 who knows, what the Dora by me still once drive let
 ‘I don't know whether Dora will let me drive her again.’

In our Saurian material there are three translation tasks with indirect polar interrogatives, cf. (23), (24), (25). In the translations *be(i)n(e)* is predominant, only two speakers (ART, GRA) use *bo*, whereby one (ART) corrects himself once to *ben* (cf. 22):

- (23) B22. La mamma mi ha chiesto se sei andato a scuola oggi.
 LAR: De mueter otber gevörschet **ben** de pist gean in der schuele haite
 AUG: De mueter otmi gevörschet **ben** 's de pist geben in der schuele haite

ART: De mueter otmi gevroget **bo** 's i pin geben in de schuele haite (poi
dice: "**ben** de pist geben in de schuele haite)

ERM: De mueter otmi gevroget **ben** de pist gean in der schuele haite

GRA: De mueter otmi gevörschet **bo** de pist gean in de schuele haite

The mother has-me asked when/where that you are gone in the school
today

'Mother asked me if you went to school today.'

- (24) E11. Mi ha chiesto se tua mamma ha ancora la tosse

LAR: Ar other gevörschet **ben** 's daina mueter ot nou de hueste

AUG: Ar otmi gevörschet **bein** 's de daina mueter ot nou de hueste

ART: 'S otmi gevroget **bo** 's daina mueter ot nou de hueste

He/it has-me asked when/where that your mother has still the cough

'He asked me if your mother still had a cough.'

- (25) E2. (Gianni) Mi ha chiesto se potete aiutarlo a riparare la macchina

LAR: Ar other gevörschet **ben** der meigetme helfn de maschine ze richtan

AUG: Ar otmi gevörschet **ben** der meiget helfnme de makina za richtan

ART: Der Gianni otmi gevroget **bo** de meigest helfn richtn de makina

GRA: Ar otmi gevörschet **bo** de meist helfnme richtn de makina

He/the G. has-me asked when/where he/you may help-me the car to fix

'He/the G. asked me if he/you could help me fix the car.'

Saurian thus differs from Standard German and most German dialects in that it does not use *ob* for indirect polar interrogatives, but *be(i)n(e)*, a complementizer that is identical to the conditional complementizer. The direction of the development is clear: from conditional to interrogative. The complementizer *be(i)n(e)* was used in conditional clauses before it came into use in interrogative clauses. *Wenn* as interrogative complementizer is also attested for other Bavarian language islands in Northern Italy: Mòcheno (*S kloa Bersntoler Beirterpuach*, s.v. *benn*), Sappada

(Benedetti & Krater 2006: s.v. *benn1*), and Cimbrian (Nicolussi Golo & Nicolussi 2014: 42, s.v. *be*).

The second complementizer in Saurian applied to introduce indirect polar interrogatives is *bo* (lit. ‘where’). Only two of our informants (ART, GRA) use it, but the dictionary by Denison & Grassegger (2007: 27, s.v. *bo-(a)s*) also lists this usage. There is at least one other Bavarian language island in Northern Italy where *wo* is used as interrogative complementizer: in Sappada, indirect polar interrogatives can be introduced by *wo* (cf. Benedetti & Krater 2006: 108, s.v. *boo*) (or by *benn*, s. above). Denison & Grassegger (2007: 27) refer to Lexer (1862) for the Carinthian, while Lexer himself refers to Schmeller. The interrogative complementizer *wo* thus seems to have been more widespread in Bavarian (in the past).

According to Schmeller (II, 828), the Bavarian complementizer *wo*, which developed from the *wh*-pronoun *wo* ‘where’,¹⁰ has three functions: it introduces relative clauses (see also Weiß 2004, 2013, Brandner & Bräuning 2013), temporal clauses (equivalent to German *als*, *da* ‘as’), or indirect polar interrogatives. As mentioned above, Schmeller (II, 828) assumes that *wo* ‘whether’ ‘perhaps originated from *wàs*’ (“vielleicht aus *wàs* entstanden”) (on *wàs* ‘whether’ cf. Schmeller II, 1016), for which, however, there is no evidence.¹¹ We assume that polar interrogative *wo* emerged from conditional *wo* thereby representing the same developmental path as *wenn* (cf. section 5). In addition to the functions mentioned by Schmeller, *wo* has been used as “konjunktion der bedingung” (DWb, s.v. *wo*, see also Behaghel 1928: 350) (‘conjunction of the condition’) at least since the 16th century. (26) is an example of the conditional use given by the DWb:

- (26) **wo** er nit kem, müst er zu busz (H. Sachs w. 6, 145)
 if he not come, must he to penance
 ‘if he does not come, he must do penance’

¹⁰ But cf. Brandner & Bräuning (2013) for a critical evaluation of the standard diachronic account for the relative complementizer *wo*.

¹¹ On *wàs* ‘whether’ cf. Schmeller II, 1016.

Since *wo* has been used as a conditional conjunction since Early High German times, it is more plausible to assume that this function formed the basis for its further development into an interrogative conjunction.

4 THE CONDITIONAL-TO-INTERROGATIVE CYCLE

4.1 THE CYCLE

In the preceding sections, we have presented cases from the history of German where a conditional subordinator developed into an interrogative subordinator. In OHG, both conditional clauses and indirect polar interrogatives shared the same subordinator *oba/ibu* ‘if/whether’, which was replaced in its conditional function by *wenn* (lit. ‘when’) in MHG. This situation is still found in NHG, that is, in Standard German and in most of the modern dialects. In a few dialects, however, the development has gone further and *ob* ‘whether’ has been replaced also in its interrogative function. This has been the case in some Bavarian varieties and in Saurian where the conditional subordinators *wenn* (lit. ‘when’) and *wo* (lit. ‘where’) replaced the interrogative subordinator *ob* ‘whether’. In these varieties, conditional clauses and indirect polar interrogatives are thus introduced by the same subordinator. This means that the initial state (i.e., a common subordinator) as it existed in OHG has been reached again. This is therefore a cyclical development. Language change often follows the form of a cycle (cf. van Gelderen 2009, 2013, 2016; Breitbarth et al. 2019; Weiß 2024): “A variant A is replaced by a variant B that eventually gets replaced by a variant C, which resembles the original variant A in some respect” (Weiß 2024: 22). The most famous cycle is probably the Jespersen Cycle that generates new negators (cf. Jäger 2008 for German), but there are many others like the comparative cycle (Jäger 2018) or the pronominal cycle (Weiß 2015). The conditional-to-interrogative cycle is a new one that has not been

described so far (as far as we know). The conditional-to-interrogative cycle in German comprises three stages (cf. Fig. 3): at stage I, there is a common subordinator for conditional clauses and indirect polar interrogatives (*oba/ibu* ‘if/whether’); at stage II, a new subordinator for conditional clauses emerged (*wenn* ‘when’ or *wo* ‘where’, respectively) that at stage III, replaces the interrogative subordinator. Stage I and III are similar in that they exhibit a common subordinator for both functions.

	conditional	interrogative
stage I	OB	OB
stage II	WENN/WO	OB
stage III	WENN/WO	WENN/WO

Fig. 3: The conditional-to-interrogative cycle

The conditional-to-interrogative cycle thus essentially involves two diachronic developments, i.e., the innovation of a conditional subordinator and the change from a conditional subordinator into an interrogative one.

4.2 FROM RELATIVE ADVERB TO CONDITIONAL SUBORDINATOR

Regarding the innovation of conditional subordinators, there seems to be a diachronic pathway that has figured prominently in the history of German, namely the development of conditional complementizers from relative adverbs in free-choice relative clauses. Behaghel (1928) argues that the ‘conjunctions’ *wenn* and *wo* go back to *sō (h)wanne sō* (cf. (27) and (28) for an Old and Middle High German example) and to *sō (h)wār(a) sō* as in (29) below. Structures of the type ‘*sō wh-pronoun/adverb sō*’ were used to introduce free-choice relatives, both nominal and adverbial ones in all West-Germanic varieties.

(27) **So huuanne so** dhu dhina daga arfullis, dhazs dhu faris zi dhinem fordhrom,

so when so you your days fulfill that you go to your ancestors

ih aruuehhu dhinen samun after dhir

I awake your seed after you

‘When you fulfill your days so that you go to your fathers, I will raise up
you seed after you’

(DDD-AD-Isidor_1.2 > I_DeFide_9)

(28) **swenne** iwer starke vînde ze helfe möhten hân drîzec tûsent degne,

when(ever) your strong foes to assistance might.SUBJ have three thousand
warriors

sô woldich si bestân,

SO wanted.SUBJ-I them bestead

‘When/If your mighty enemies had thirty thousand warriors to aid them, I
would still withstand them (= the king’s warriors)’

(Nib. 374,3, cited from BMZ, vol 3, column 503b)

Interpretatively, these seem to be parallel ‘ever’ free relatives (Šimík 2020). If preposed, OHG and MHG free relatives were resumed by a (sometimes silent) correlative pronoun or pro-adverb in the main clause. In the MHG example in (28), this adverb is *sô*, which was semantically underspecified and could occur with all types of adverbial clauses including conditional ones where *so* is still possible even in modern Standard German (Axel-Tober 2023). The whole structure has been argued to be correlative relativization (Axel-Tober 2023), which is a major relativization strategy cross-linguistically (cf. de Vries 2002) and was a prominent characteristic of old Germanic subordination (Kiparsky 1995, Axel-Tober 2012). In OHG, *sô wār sô*-relatives could be used with the local adverb *thār/dār* ‘there’ as a correlative (29a), or with *sô* ‘so’ (29b). Correlative *sô* seems to fit an observation by Behaghel (1928) that local free relative clauses may come close to a conditional-like interpretation.

(29) a. **so uuâr sô** dû bist . **dâr** ist ér.

so where so you are there is he

‘Wherever you are, he is there.’

(Notker: Psalter, Psalm 74,267,26f.; cited from Lühr 1998: 267, ex. 19;
English glosses and translation added)

- b. **So uuare sose** ih cherte minen zoum, **so** rado nami dus goun:

so where so-so I turned my rein so took you-it care’

‘Wherever I turned my reins, that's what you were always looking for.’

(Psalm 138,6; cited from Lühr 1998: 267, ex. 21)

In the following example from Otfrid, two local relative clauses are post-posed. The first relative is a free *d*-relative introduced by *thār* ‘there’,¹² the second a *wh*-relative introduced by *sō wār*.

- (30) ni mag diufal thara ingegin sin, **thar** ir ginennet namon min (O. V, 16, 33)

NEG may devil there opposite be there you name name my

so war ir es biginnet, ir widar imo ringet

so where you it.GEN begin you against him wrestle

‘The devil may not be vis-à-vis you (at the place) where you utter my name.

Wherever (if?) you do that, you will wrestle against him’

(cited from Behaghel 1928: 350; glosses and translation added)

According to Grimm’s dictionary (DWb vol. 30, column 903), *wo* as a “conditional conjunction” begins to be robustly attested in the 14th century, witnessed by the fact that counterfactual *wo*-clauses in the preterite subjunctive start to occur in that period (Behaghel 1928: 350). Conditional *wo* was common until the 17th century and was pushed back in written (Standard) language during the 18th century.

¹² *D*-pronoun free relatives (*ther*, *thiu*, *thaz* or *thar*, *thō*, *thanne*) were a further type and these could also have a universal- or definite-like interpretation (Axel-Tober 2012: 296).

Behaghel cites the OHG example from Otfrid to illustrate that free *wh*-relatives of the *sō-wār-sō*-type may come close to a conditional interpretation:¹³

Aus der greifbaren Örtlichkeit kann aber die geistige hervorgehen; *wo* bezeichnet dann die verschiedenen Lagen, für die der Hauptsatz gilt, und der Nebensatz kommt der allgemeinen Bedingung nahe (eine Entwicklung, die auch *thar* durchgemacht hat).

‘From the tangible locality, however, the conceptual one can arise; *wo* then denotes the different situations to which the main clause applies, and the subordinate clause comes close to the general condition (a development which has also *thar* undergone).’

Wenn and *wo* must thus have been ultimately subject to a change from a temporal or local adverb into a complementizer. The source was a *wh*-adverb preceded and followed by *sō* in free-choice relative clauses (cf. Weiß 2016 or Jäger 2018).¹⁴ The free-choice construction¹⁵ still exists today in Standard German and shows *wann* and *wo* as relative adverbs:¹⁶

- (31) a. Wann immer sie ihn fragte, begleitete Hans seine Nichte Maria.
 ‘Whenever she asked him, Hans accompanied Maria, his niece.’
 b. Elif sucht eine Wohnung, wo (auch immer) ihre Freundin einen Job findet.
 ‘Elif is looking for an apartment where (whatever) her friend can find a job.’

¹³ More recently, formal semantic accounts derive the conditional interpretation of free relatives by means of a covert universal quantifier binding the relative pronoun (von Stechow 2011, Caponigro 2004).

¹⁴ Lühr (1998) goes even so far as to argue that the *sō* to the right of the *wh*-phrase was the universal adverb with the meaning ‘ever’, but this analysis is problematic (Weiß 2016).

¹⁵ Free-choice give rise to a universal-like and definite-like interpretation in many languages (Šimík 2020). In modern German, *wann* can only be used in universal-like free relatives as in (31a) above (Axel-Tober 2012: 297) and has to be accompanied by the adverb *immer* ‘always, ever’. The definite-like reading seems to be not possible, cf. (i). *Wo* has both a universal-like reading (cf. (31b) above) and a definite-like one, cf. (ii).

(i) *Wann sie eingeschult wurde, begleitete Hans seine Nichte Maria.
 ‘When she started school, Hans accompanied his niece Maria.’

(Axel-Tober 2012: 297, ex. 49a,b)

(ii) Elif wohnt, wo ihr Eltern wohnen.
 ‘Elif lives (at the same place) where her parents live.’

¹⁶ In contrast to *wo*, *wann* is no longer in use as a relative adverb in headed relatives in Standard German (cf. ?* *Das letzte Mal, wann wir uns trafen*).

In Standard German, the umlaut discriminates between the adverb *wann* (relative, interrogative) and the complementizer *wenn*. The second *sō* in *sō wanne sō* is often dropped already in OHG. In late OHG (Notker), we find unverbated forms with the first *sō* (*sowanne/sowenne*). In MHG this is reduced to *swanne/swenne*, cf. (15) and (28) above. During the 14th century the short forms *wenne* or *wen* already prevail in some scripts (DWb, vol. 29, column 51).

For the development from the *wh*-adverbs *wann/wo* to the subordinating conjunctions *wenn/wo* in adverbial clauses, we adopt Jäger's (2018: 478) analysis for comparative *wie* 'how' in (32):

(32) $[_{CP} \dots [_{CP} sō wio [_{C^o} sō]] > [_{CP} \dots [_{CP} [_{C^o} (s)wie] \dots]]$

We therefore assume that this reanalysis followed the Head Preference Principle (van Gelderen 2004: 18).¹⁷ As a result of the reanalysis, *wann/wenn* and *wo* were recategorized as complementizers, i.e. as C^o -elements in traditional terms or as $Force^o$ -elements in a Rizgian-style analysis. More precisely, this must have given rise to a lexical split. The *wh*-adverb (= *wann* in Standard German) was retained, but additionally the temporal complementizer *wann/wenn* was innovated as a new lexical item. Unfortunately, it cannot be determined in the historical corpus data when the Spec-to-Head reanalysis must have taken place.¹⁸ To date, there are no in-

¹⁷ This essentially involved the loss of a movement structure and the base generation of the emerging complementizer in a C^o -position in the diachronic target (cf. van Gelderen 2009 on 'Spec-to-Head' and 'Merge over Move' in cyclical change).

¹⁸ For *when*-type subordinators in modern Germanic, there is an alternative (synchronic) analysis. Even though they are usually called 'complementizers' in the theoretical literature, they could potentially be analyzed as lexicalizations of a relative operator in SpecC. For English, there is a strand of research since Geis (1978) which analyses *when*-clauses as relative clauses (cf. Haegeman 2008 for an overview of the literature). A major argument in favor of such an analysis is the availability of a 'low' reading in examples like *I saw Mary in New York when she claimed that she would leave* (= 'I saw her at the time of her presumed departure') alongside the 'high' construal reading ('I saw her at the time she made the claim'). The low reading can be derived if a relative operator is involved that cannot only originate in the higher, but also in the embedded clause:

(i) low construal: 'I saw Mary in New York [_{CP} when_i [_{TP} she claimed [_{CP} t_i that [_{TP} she would leave t_i]]]]

(Haegeman 2008: 389, ex. (5), based on observations by Larson 1987)

depth and conclusive investigations of the interpretative developments of *wann/wenn* and *wo*. *Wenn* was used sporadically to express more general temporal interpretations in Early New High German, including anteriority (‘after that’) and simultaneity in the past (cf. the examples in Ebert et al. 1993: 414, § S 173). This is no longer possible in Standard German, where temporal *wenn* can only be used in sentences referring to recurring events or processes in the past. *Wo* functions as an adverbial complementizer with a temporal meaning in certain dialects (Behaghel 1928: 350) and it may be used with a causal interpretation in Standard German. As we have shown above, there is an interpretative closeness between the temporal or local free-choice relative and a conditional.¹⁹ This would suggest that the semantic reanalysis into a conditional went hand in hand with the syntactic reanalysis of the adverb in the free-choice relative construction into a complementizer.²⁰

The corresponding German example with *als*, however, only allows for a high construal (*Ich habe Maria in New York getroffen, als sie behauptete, dass sie weggehen würde*), and this is one of the reasons why the relative-clause analysis has not figured so prominently for German. As can be seen in (i), Haegeman (2008) analyzes English *when* as the lexicalization of the relative operator. This type of relative-clause analysis of temporal adverbial clauses could, however, also involve a silent operator moved to SpecCP and *when* could then still be a complementizer merged in C° (cf. Axel-Tober 2012: 328 on Middle High German *dō*). To complicate things further, English *if*-conditionals do not allow for a low construal either and are still analyzed as free relatives over possible worlds in Bhatt & Pancheva (2006). Under this type of analysis, conditional *wenn* in German could be treated as an operator just the same.

As it is generally the case with diachronic scenarios, the actual analysis of the purported change is, of course, dependent on the synchronic analysis of the target. Since we cannot settle the issue in this diachronic paper which type of analysis is most appropriate for Standard clauses *wenn*-clauses and their dialectal counterparts, we will stick to the traditional analysis here that *wenn* is a complementizer in temporal and conditional clauses.

¹⁹ Alternatively, Hara (2022) proposes that conditional *wenn* developed from a causal meaning along the “single path of temporal-causal-conditional, (h)*wanta* > *wanta* + *swenne/swanne* > *wenn*”. She states that “it has been vaguely presumed that the shift from causal to conditional occurred between 1400–1550 (Ebert et al. 1993, 473, Paul 2007, 422)”. However, the reference grammars that she gives here as sources for this claim do not actually discuss this purported development from causal to conditional. She furthermore cites Gagel (2017), who speculates that causal *wanta* (from OHG *hwanta* > Mhd *wande*) may have merged with free-choice *swenne/swanne* in the 15th century. Gagel does indeed seem to argue that Early New High German *wann* (with umlaut) may have a conditional interpretation. However, in his account, this does not develop from the older causal meaning through regular mechanisms of meaning change, but through the merge with *wenn/wenne*. Also, it is mostly unclear which meaning components are really part of the semantics and which ones are pragmatic inferences (Gagel 2017: 114).

²⁰ Additionally, there must have been changes in the external syntax, but this is beyond the scope of the present paper.

4.3 FROM CONDITIONAL TO INTERROGATIVE SUBORDINATOR

The second pathway relates to the diachronic shift conditional subordinator > interrogative subordinator. From a semantic perspective, this is more unexpected than the development of relative adverbs in free relative constructions into conditionals. In a recent formal-semantic discussion of the ‘interrogative-conditional’ link from a synchronic and diachronic perspective, Iatridou & Zeijlstra, sub., p. 3) come to the conclusion that “the semantics of conditionals and the semantics of questions are quite far apart” (Iatridou & Zeijlstra sub., p. 3). For conditionals, they refer to the problem that in the restrictor theory, which has become one of the most prominent, if not the standard theory, the conditional subordinator is semantically empty. Its only contribution is to mark the restriction to an (often covert) operator, cf. (33) for an example based on Lewis (1998: 10, ex. 32).

- (33) a. Often/Always/Sometimes if it is raining my roof leaks.
 b. If it is raining my roof leaks.

In indirect interrogatives, by contrast, *if* and its cross-linguistic counterparts contributes a meaning through which the interrogative clause comes to denote the set of its possible answers (e.g. Hamblin 1976), i.e. p and $\neg p$ in yes/no interrogatives.

- (34) a. $[[\text{if it is raining}]] = \{\text{it is raining}, \neg \text{it is raining}\}$
 b. $[[\text{if}]] = \lambda p. \lambda q. q = p \vee q = \neg p$

In these standard approaches, there is thus no semantic closeness between conditional and interrogative complementizers and it is not conceivable which kind of meaning-change mechanism could explain the diachronic evolution of an interrogative from a conditional complementizer.

However, there are alternative theories which posit a synchronic interpretive affinity and could thus potentially be used to account for diachronic mechanisms of change in this area. Iatridou & Zeijlstra (sub.) refer to the suppositional theory by Starr (2014) where the conditional protasis serves to restrict the context of the apodosis assertion. The “if-clause contributes a supposition under which the consequent alone is accepted or asserted” (ibid. p. 1). In this theory, not only the homonymy of interrogative and conditional subordinators but also structural parallels between interrogative and conditionals (cf. section 5 on V-to-C conditionals) fall out immediately, cf. the following examples and paraphrases based on Starr (2014):

- (35) a. Raise the Yes/No question “p?” ($\{p, \neg p\}$)
 b. Take the positive answer as a highlight (so “suppose p”)
 c. Then p should entail q

- (36) If it is raining, Elif’s roof leaks.
 Suppose that we are wondering if it is raining ...
 ... and we focus on worlds where it is ...
 Then we will find out that Elif’s roof leaks.

However, Iatridou & Zeijlstra confront the suppositional theory with a number of objections. For example, it seems to be unable to account for counterfactual, post-posed or factual conditionals. The author therefore conclude that Starr’s question-based semantics is synchronically untenable, but it offers a potential explanation for a diachronic spread from conditional to interrogative meanings (and vice-versa) through what they call semantic spread „facilitated by the similarity in usage conditions between polar questions and conditional clauses” (ibid., p. 27) . They exemplify this by means of a scenario for the potential development of Latin *si*, which was first used with verbs of expectation and doubt, before it developed into

a general subordinator for indirect interrogatives also with verbs such as *rogare* ‘ask’:

- (37) *Exspectabamque, si quid de eo ad me scriberes.*
 wait.IND.IMPF.1SG-and if something about this to me write.SUBJ.IMPF.2SG
 ‘I was waiting, in case you would write me something’
 (Iatridou & Zeijlstra sub. ex. (46) with reference to Bodelot 2013)

This example is still clearly conditional as given in the translation, but it can also receive an interrogative interpretation ‘I was waiting/wondering whether you would write me something’. However, while this scenario is convincing for the development of Latin *si*, it should make explicit the properties of the matrix predicates that give rise to potential contexts with such similar usage conditions. The etymological study by Lühr (2007) contains an observation on the semantic relation between polar interrogatives and conditional clauses in German ‘minimal pairs’ such as the following:

- (38) a. Hans beobachtet, **wenn** ein Unfall passiert.
 ‘Hans will observe (= watch/pay attention) it, if an accident happens.’
 b. Hans beobachtet, **ob** ein Unfall passiert.
 ‘Hans is observing (= watching/paying attention) whether an accident is going to happen.’

She argues that the polar interrogative denotes a pair of propositions, i.e. the embedded proposition and the complementary proposition (‘Hans is watching whether an accident is going to happen or whether an accident is not going to happen’). In the construction with the conditional clause, the negated proposition is ruled out, but in both cases, interrogative and conditional, the subordinate clause can be argued to relate to the preconditions that have to be fulfilled for something to be the case.

Lühr doesn't discuss which types of matrix predicates give rise to this interpretive relatedness. A prerequisite seems to be that they need to subcategorize for a declarative besides an interrogative clause as exemplified for the verb *beobachten* 'observe' in (39).

(39) Hans beobachtet, **dass** ein Unfall passiert.

'Hans is observing that an accident is happening.'

Lahiri (2002) calls the predicates that allow for both types of complementation 'responsive' predicates as opposed to 'rogative' predicates that only take interrogative complements. Many German perception verbs (*beobachten* 'observe, watch', *sehen* 'see', *wahrnehmen* 'perceive', *bemerken* 'notice', *hören* 'hear', *fühlen* 'feels', *schmecken* 'taste'), reportative verbs (*erzählen* 'tell', *berichten* 'report', *informieren* 'inform', *darlegen* 'demonstrate' etc.) and the verb *wissen* 'to know' fulfill this requirement, cf. (40) for further examples.

(40) a. Hans berichtet, **ob/dass/wenn** ein Unfall passiert.

'Hans will report it whether/that/if an accident happens.'

b. Hans weiß (es), **ob/dass/wenn** ein Unfall passiert ist.

'Hans knows it whether/that/if an accident has happened.'

Interestingly, the declarative and interrogative uses of responsive predicates seem to quite generally be encoded by the same lexical item not just in German and English, but also in typologically unrelated languages (cf. Elliot et al. 2017 for further discussion and examples). With this group of matrix predicates conditional clauses and polar interrogatives have 'similar usage conditions' to put in Iatridou's & Zeijlstra's (sub.) terms. They could thus have figured in potential 'bridging contexts' for a diachronic reanalysis of a conditional into an interrogative construction.

The change must have included a semantic reanalysis of the subordinator along with several syntactic changes. As mentioned in section 2, in the case of German *ob*, it cannot be established whether the interrogative meaning is derived from the conditional one. With the *wenn*- and *wo*-clauses the conditional use is clearly older. Their developments into interrogatives probably took place much later, during the New High German period, but only in certain dialects. Unfortunately, we do not have enough data on the historical precursors of those dialects so that we cannot pinpoint the timing of the change. The constructed modern Standard German minimal pairs with *ob* provide a synchronic window into the diachronic pathway that we hypothesize here. Interrogative complementizers are standardly analyzed as C° elements (but see Kayne 1991: 666 for a phrasal analysis of English *whether*). Under the perspective of a Rizgian-style analysis, the innovation of the interrogative complementizer in Saurian, can be analyzed as downward reanalysis from ForceP to IntP (cf. 27a, b) which is an instance of Weiß's (2019) Early Merge Principle. Note that the same analysis can be applied to the development of *wo*, which evolved from locative to conditional to interrogative.

- (41) a. [_{ForceP} *be(i)n(e)*] [_{IntP} [_{TopP} [_{FocP} [_{FinP} (*a*)s ...] →
 b. [_{ForceP} [_{IntP} *be(i)n(e)*] [_{TopP} [_{FocP} [_{FinP} (*a*)s ...]

This change must have led to a further lexical split. The older conditional subordinator (also a complementizer according to our analysis) was retained in the respective dialects even after the 'homonymous' interrogative complementizer had been innovated (= phase II of the cycle).

In the external syntax, the change from conditional into interrogative would involve a reanalysis of an adjunct into a complement clause, a type of change that is also attested independently (cf. Axel-Tober 2017 on Germanic *that*-type clauses and É. Kiss 2023 on Hungarian *hogy*-clauses). A further well-known example is the development of complement clause after emotive predicates from causal adjunct

clauses (cf. the English paraphrase: *Timosa was angry because you left* > *Timosa was angry that you left*).^{21 22}

5 FROM INTERROGATIVE TO CONDITIONAL?

The semantic affinity between conditionals and interrogatives after certain matrix predicates suggests that there could not also be a development from conditional into interrogative as discussed in the previous sections, but also the reverse, i.e. a shift from interrogative to conditional. Under the perspective of similarity of use conditions and semantic spread, Iatridou & Zeijlstra (sub. P. 27) conclude that “one may also expect reverse scenarios where a conditional construction semantically spreads into an interrogative construction.” As empirical evidence for such a diachronic scenario, they refer to the phenomenon of Germanic V1 conditionals. In modern English, in contrast to Old English or OHG, these are restricted to counterfactual apodoses with the verbs *have*, *be* or *should*:

(42) Should the sun shine, we could go for a walk.

Such conditionals are often analyzed as grammaticalizations of discourse sequences as illustrated in (43):

(43) A: Scheint die Sonne? = S1

²¹ Cf. Nicholas (1998) for an account of how the Greek declarative *pu*-complementizer may have resulted from a reanalysis of a causal precursor. Outside Indo-European, similar developments have been reported for Semitic (e.g. Deutscher 2000 on Akkadian *kīma*; cf. Axel-Tober & Weiß to appear for further examples).

²² In synchronic theoretical approaches, the classical view that so-called object clauses are syntactically complements and semantically propositional arguments of their matrix predicates has been challenged recently. Analyses have been advocated that treat such clauses as relative clauses (Arsenijević 2009; Kayne 2014), in some work this is also explicitly extended to indirect interrogative clauses (Arsenijević 2009). This would also have implications for the putative diachronic processes. The shift from conditional to interrogative would then potentially involve a continuation of an adjunct-like structure in the external syntax.

shines the.nom sun

B: Ja.

yes

A: So/Dann gehen wir baden. = S2

so/ then go.1pl we bathe

‘A: Is the sun shining? B. Yes. A: Then, we’ll go for a swim.’

(based on van den Nest 2010, 95)

Van den Nest offers a grammaticalization of this. However, he also points out that the historical record in Old English poses some challenges for such a scenario, in particular the high percentage of V1-apodoses in the present subjunctive in his data. The present subjunctive is hardly possible in direct polar interrogatives. It should also be mentioned that V1-conditionals do not behave like canonical conditionals even in languages where they are not restricted to certain verbs or verbal moods (Axel & Wöllstein 2009, Reis & Wöllstein 2010). It is thus unclear which role such structures should play in the general discussion of the conditional-interrogative link from a diachronic perspective. According to Weiß (2019), such grammaticalizations of discourse sequences, where a paratactic sequence of two independent clauses results in a hypotactic sentence structure consisting of a main clause and a subordinate clause, are ruled out on principle, and there is not a single convincing example of such a development (Weiß 2020, Axel-Tober & Weiß, accepted).

As far as complementizers are concerned, there are some mentions in the literature for a diachronic path interrogative > conditional, but largely without proper reference to concrete diachronic examples. Walkden, Li & Weisshaar (2025) even go so far as to say that “the pathway from PQP [= polar question particle; the authors] appears to be well-trodden (Heine & Kuteva 2002: 249; Hopper & Traugott 2003: 186 and literature cited there; Kuteva et al. 2019: 353–354; Iatridou & Zeijlstra to appear)” and that they “are not aware of any evidence that the reverse pathway is attested at all.” As to their latter statement, we have provided

counterevidence to this in our paper: There are uncontroversial examples for the pathway ‘conditional > interrogative complementizer’. Regarding their former claim, if one checks the literature that they refer to here, it turns out that the only paradigmatic example for ‘interrogative > conditional’ cited repeatedly is actually the (Germanic) V1 conditional structures. Heine & Kuteva (2002) and Kuteva et al. (2019) also mention Russian *esli* and its putative evolution from *est’ li* ‘is it?’. *Esli* has been in use as a conditional subordinator since the 16th century, but it is not interrogative anymore and its origin is far from clear (Tilman Berger p.c.). It will thus have to be left to further research to investigate further potential candidates for the pathway ‘interrogative > conditional’.

6 CONCLUSIONS

Our paper has dealt with a hitherto little addressed issue, namely that some languages use the same complementizer for conditional clauses and indirect polar interrogatives. This phenomenon is attested not only in Indo-European languages, but in languages from other language families as well – for example, in Gorwaa (South Cushitic, Tanzania), Maltese (Semitic), or Malagasy (Austronesian). Using German as an example, we show that there is a cyclical development here: in OHG, both clause types were introduced by the complementizer *ibu/oba* (cognate to English *if*), which was later replaced in its conditional function by *wenn* ‘when’. In some dialects (for example, in Saurian), however, there was then a further developmental step: the new conditional complementizer also entered the domain of indirect polar questions, so that, as in OHG, there is once again a common complementizer for both functions. We are therefore dealing with a conditional-interrogative cycle and the development seems to always go in one direction, from conditional to interrogative. In contrast to claims in the literature, we have not found any no reliable evidence for the reverse development from interrogative to conditional so far (as we discussed in section 5).

The phenomenon shows that there is a ‘conditional-interrogative link’ at the syntactic level, but it is not easy to reconstruct this semantically. In the standard approaches, there is no semantic closeness between conditional and interrogative complementizers, so it is not clear which kind of meaning-change could explain the diachronic evolution of an interrogative from a conditional complementizer. In section 4, we propose (following Lühr 2007) that a group of matrix predicates like *beobachten* ‘observe, watch’ or *berichten* ‘report’ may give rise to an interpretive relatedness between conditional clauses and polar interrogatives. In sentences with these predicates conditional clause and interrogative complements have ‘similar usage conditions’ and thus such constructions could function as potential ‘bridging contexts’ for a reanalysis of a conditional into an interrogative construction.

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